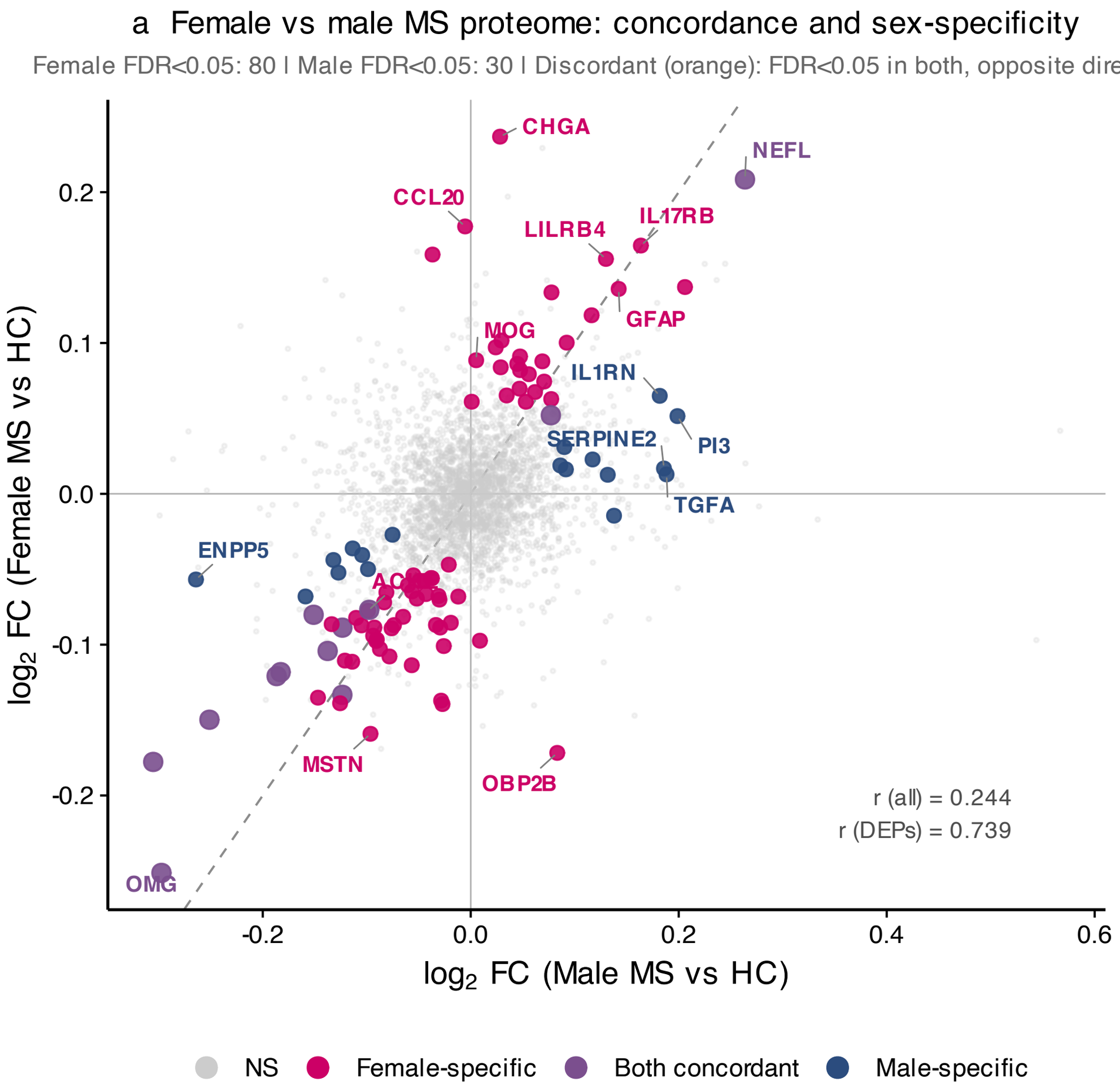
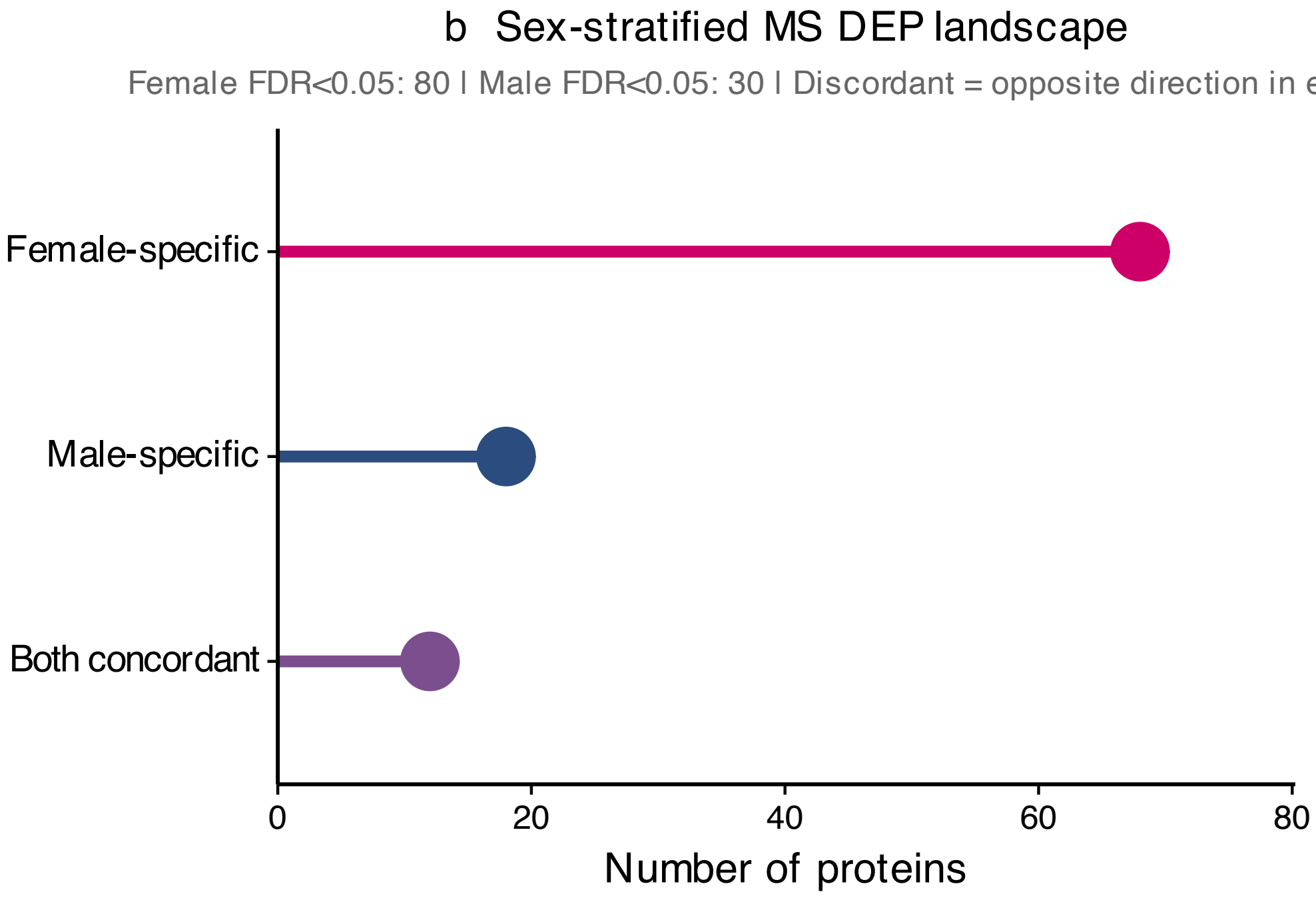
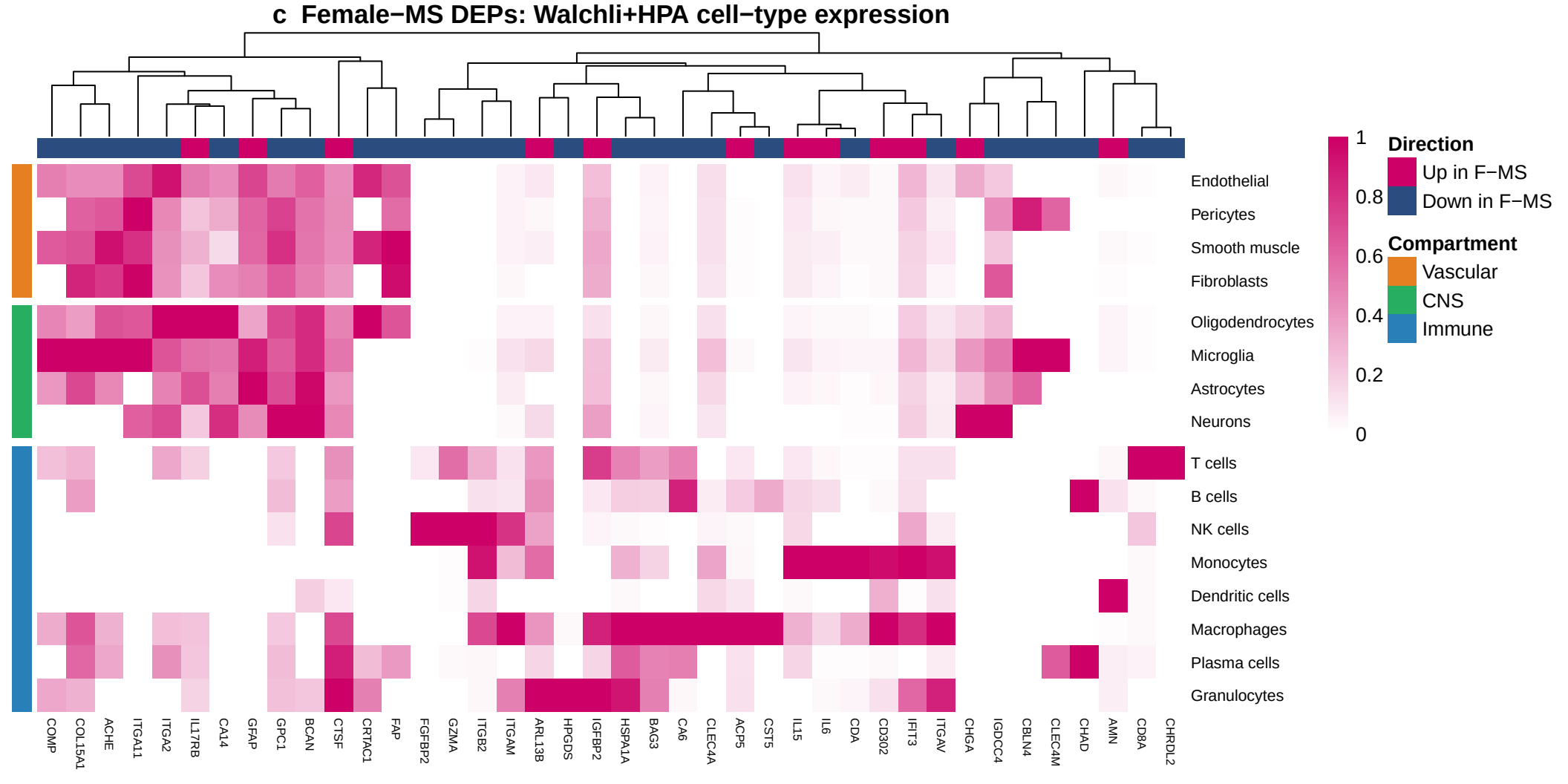
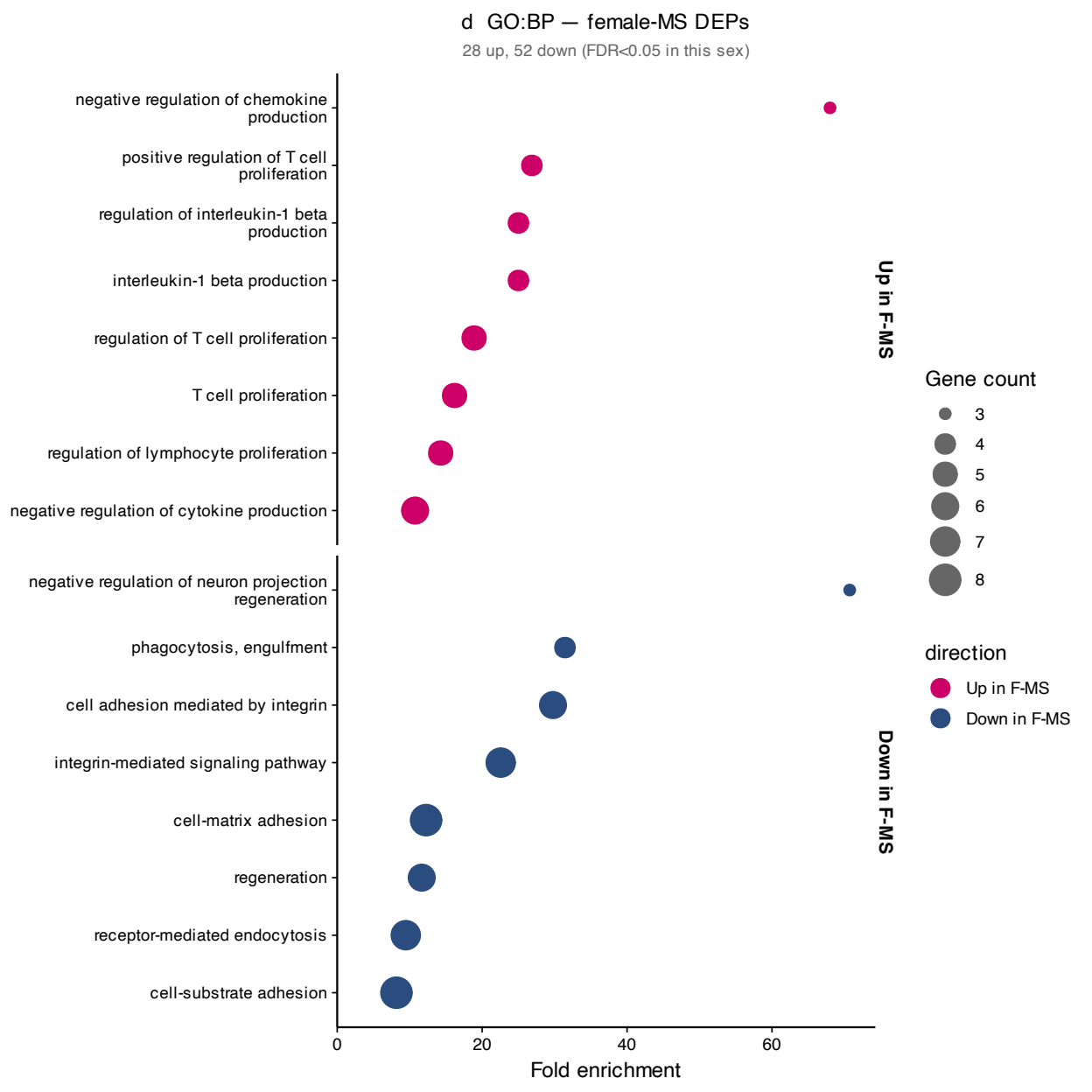
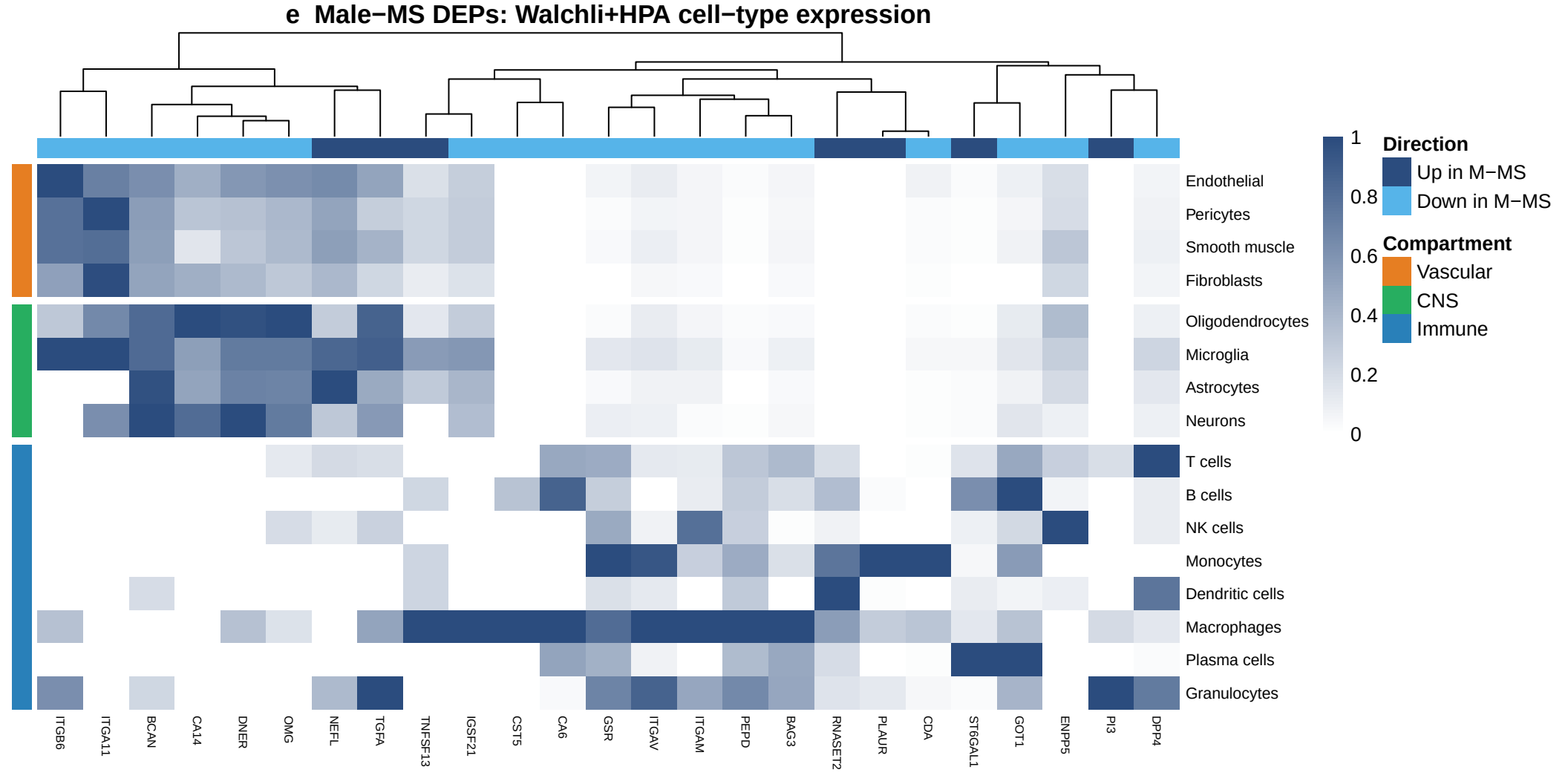
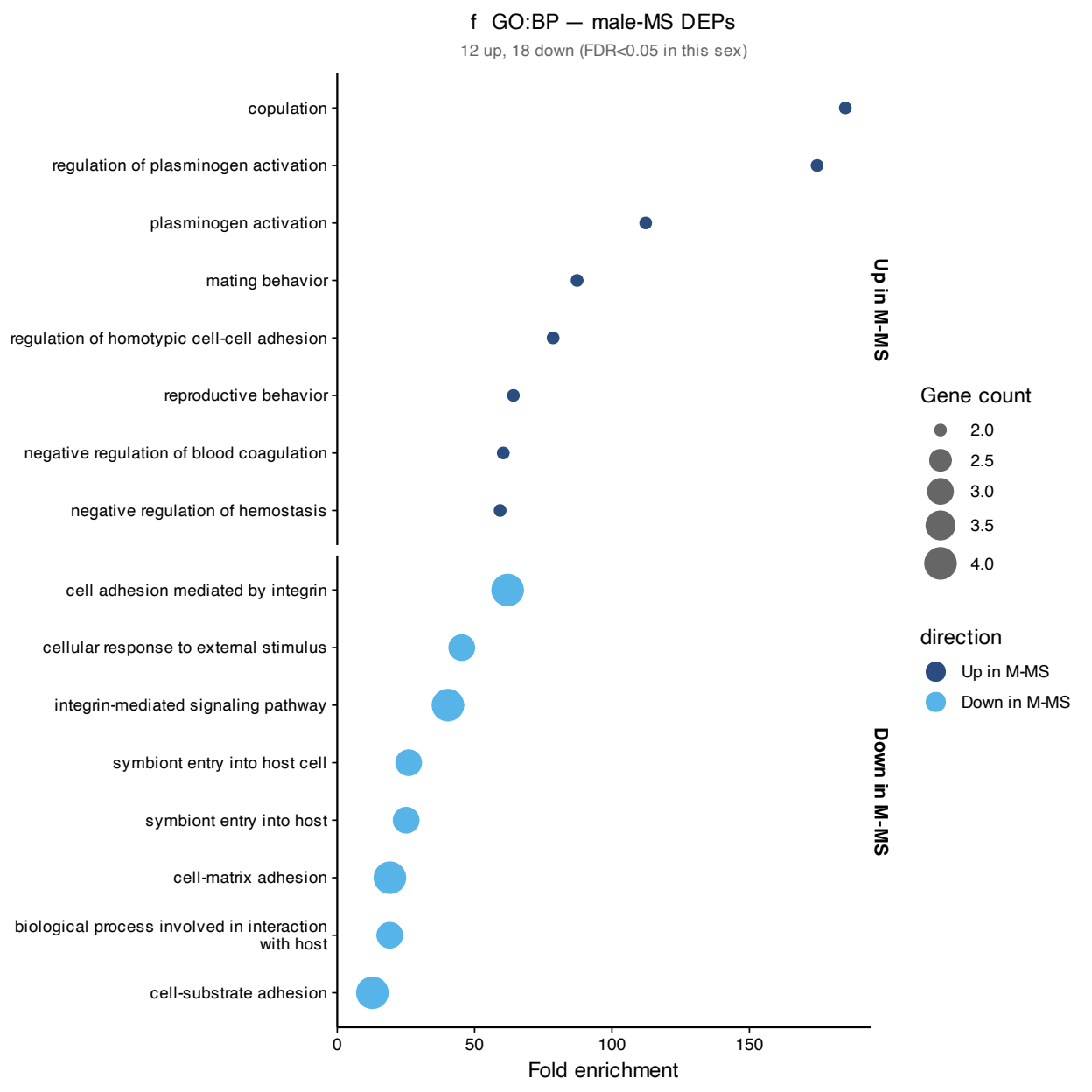

a Female vs male MS proteome: concordance and sex-specificity
Female FDR<0.05: 80 | Male FDR<0.05: 30 | Discordant (orange): FDR<0.05 in both, opposite dire
CHGA
NEFL
CCL20
IL17RB
LILRB4
GFAP
MOG
IL1RN
SERPINE2
PI3
TGFA
ENPP5
MSTN
OBP2B
OMG
r (all) = 0.244
r (DEPs) = 0.739
0.2
0.1
0.0
-0.1
-0.2
-0.2
0.0
0.2
0.4
0.6
log2 FC (Female MS vs HC)
log2 FC (Male MS vs HC)
NS
Female-specific
Both concordant
Male-specific

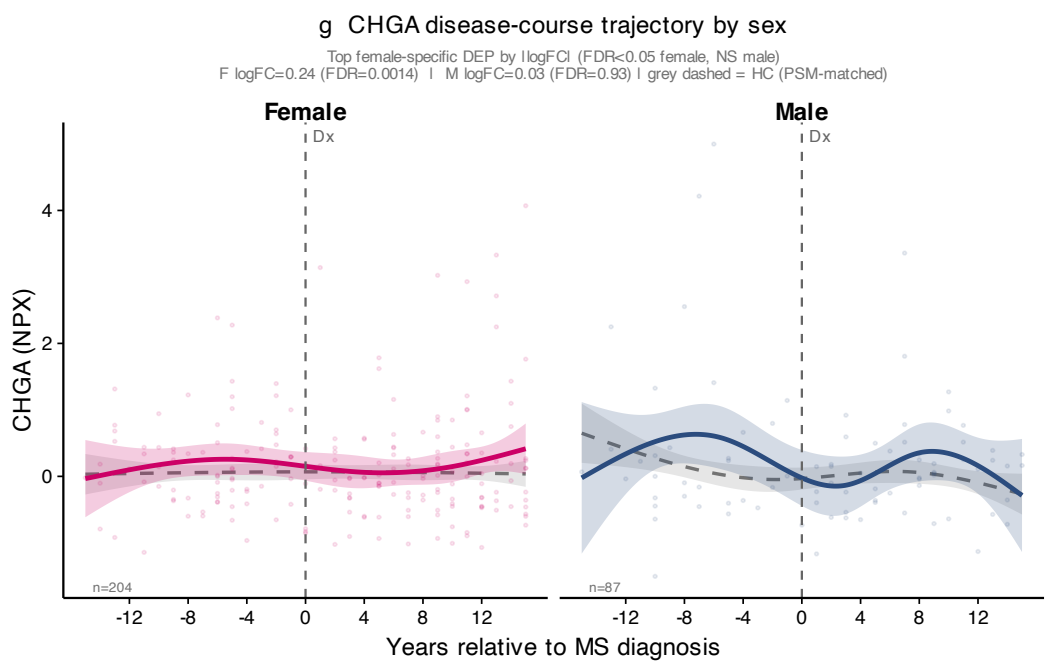

b Sex-stratified MS DEP landscape
Female FDR<0.05: 80 | Male FDR<0.05: 30 | Discordant = opposite direction in e
Female-specific
Male-specific
Both concordant
0
20
40
60
80
Number of proteins

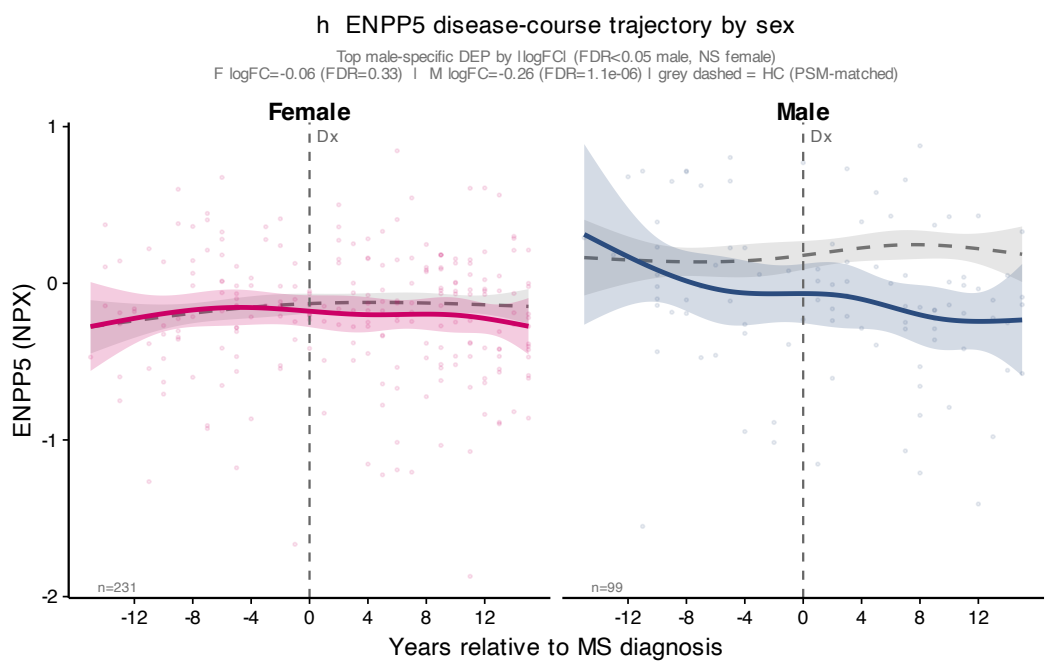

c Female−MS DEPs: Walchli+HPA cell−type expression
Endothelial
Pericytes
Smooth muscle
Fibroblasts
Oligodendrocytes
Microglia
Astrocytes
Neurons
T cells
B cells
NK cells
Monocytes
Dendritic cells
Macrophages
Plasma cells
Granulocytes
COMP
COL15A1
ACHE
ITGA11
ITGA2
IL17RB
CA14
GFAP
GPC1
BCAN
CTSF
CRTAC1
FAP
FGFBP2
GZMA
ITGB2
ITGAM
ARL13B
HPGDS
IGFBP2
HSPA1A
BAG3
CA6
CLEC4A
ACP5
CST5
IL15
IL6
CDA
CD302
IFIT3
ITGAV
CHGA
IGDCC4
CBLN4
CLEC4M
CHAD
AMN
CD8A
CHRDL2
1
0.8
0.6
0.4
0.2
0
Direction
Up in F−MS
Down in F−MS
Compartment
Vascular
CNS
Immune

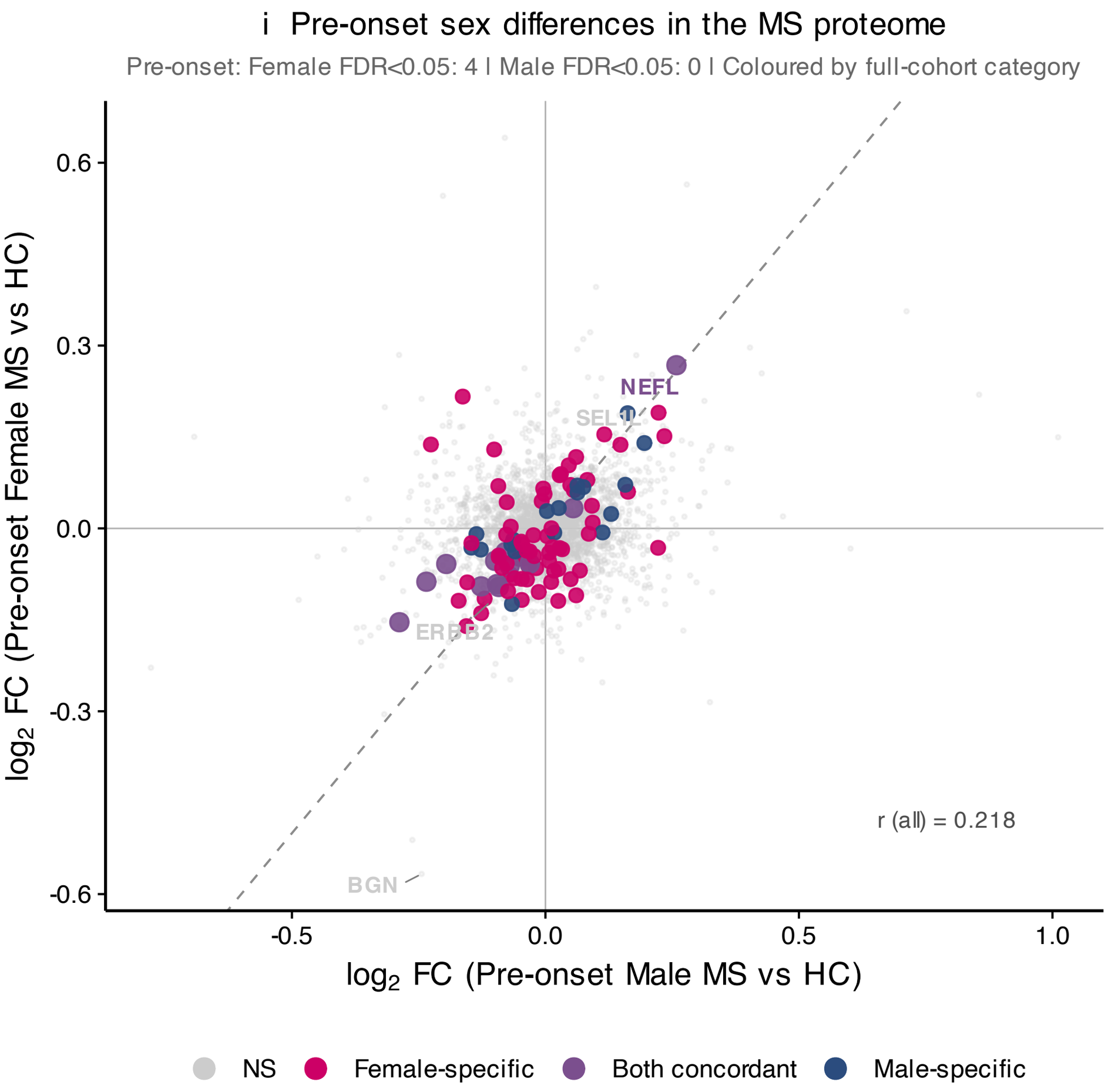

d GO:BP — female-MS DEPs
28 up, 52 down (FDR<0.05 in this sex)
negative regulation of chemokine production
positive regulation of T cell proliferation
regulation of interleukin-1 beta production
interleukin-1 beta production
regulation of T cell proliferation
T cell proliferation
regulation of lymphocyte proliferation
negative regulation of cytokine production
negative regulation of neuron projection regeneration
phagocytosis, engulfment
cell adhesion mediated by integrin
integrin-mediated signaling pathway
cell-matrix adhesion
regeneration
receptor-mediated endocytosis
cell-substrate adhesion
Up in F-MS
Down in F-MS
0
20
40
60
Fold enrichment
Gene count
3
4
5
6
7
8
direction
Up in F-MS
Down in F-MS

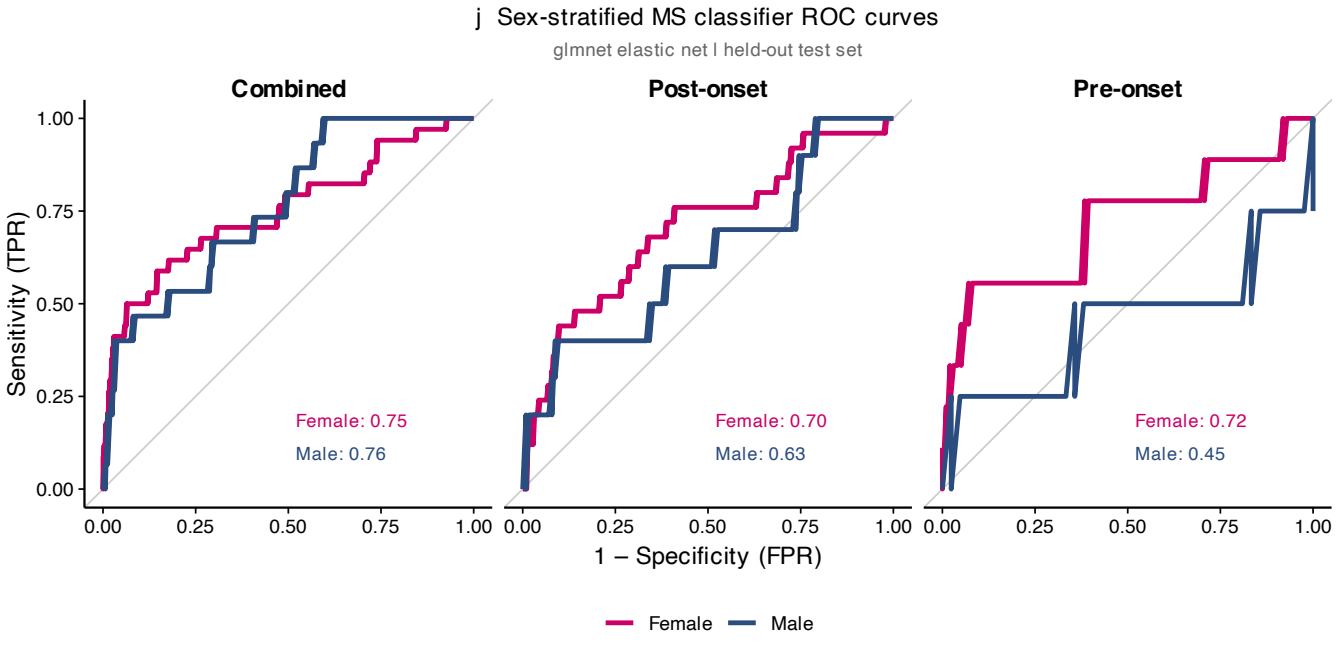

e Male−MS DEPs: Walchli+HPA cell−type expression
Endothelial
Pericytes
Smooth muscle
Fibroblasts
Oligodendrocytes
Microglia
Astrocytes
Neurons
T cells
B cells
NK cells
Monocytes
Dendritic cells
Macrophages
Plasma cells
Granulocytes
ITGB6
ITGA11
BCAN
CA14
DNER
OMG
NEFL
TGFA
TNFSF13
IGSF21
CST5
CA6
GSR
ITGAV
ITGAM
PEPD
BAG3
RNASET2
PLAUR
CDA
ST6GAL1
GOT1
ENPP5
PI3
DPP4
1
0.8
0.6
0.4
0.2
0
Direction
Up in M−MS
Down in M−MS
Compartment
Vascular
CNS
Immune


f GO:BP — male-MS DEPs
12 up, 18 down (FDR<0.05 in this sex)
copulation
regulation of plasminogen activation
plasminogen activation
mating behavior
regulation of homotypic cell-cell adhesion
reproductive behavior
negative regulation of blood coagulation
negative regulation of hemostasis
cell adhesion mediated by integrin
cellular response to external stimulus
integrin-mediated signaling pathway
symbiont entry into host cell
symbiont entry into host
cell-matrix adhesion
biological process involved in interaction with host
cell-substrate adhesion
Up in M-MS
Down in M-MS
0
50
100
150
Fold enrichment
Gene count
2.0
2.5
3.0
3.5
4.0
direction
Up in M-MS
Down in M-MS


g CHGA disease-course trajectory by sex
Top female-specific DEP by |logFC| (FDR<0.05 female, NS male)
F logFC=0.24 (FDR=0.0014) | M logFC=0.03 (FDR=0.93) | grey dashed = HC (PSM-matched)
Female
Male
Dx
Dx
CHGA (NPX)
4
2
0
n=204
n=87
-12
-8
-4
0
4
8
12
Years relative to MS diagnosis


h ENPP5 disease-course trajectory by sex
Top male-specific DEP by |logFC| (FDR<0.05 male, NS female)
F logFC=-0.06 (FDR=0.33) | M logFC=-0.26 (FDR=1.1e-06) | grey dashed = HC (PSM-matched)
Female
Male
Dx
Dx
ENPP5 (NPX)
1
0
-1
-2
n=231
n=99
-12
-8
-4
0
4
8
12
Years relative to MS diagnosis


i Pre-onset sex differences in the MS proteome
Pre-onset: Female FDR<0.05: 4 | Male FDR<0.05: 0 | Coloured by full-cohort category
NEFL
SELTL
ERBB2
BGN
r (all) = 0.218
0.6
0.3
0.0
-0.3
-0.6
-0.5
0.0
0.5
1.0
log2 FC (Pre-onset Female MS vs HC)
log2 FC (Pre-onset Male MS vs HC)
NS
Female-specific
Both concordant
Male-specific


j Sex-stratified MS classifier ROC curves
glmnet elastic net | held-out test set
Combined
Post-onset
Pre-onset
Sensitivity (TPR)
1 – Specificity (FPR)
0.00
0.25
0.50
0.75
1.00
Female: 0.75
Male: 0.76
Female: 0.70
Male: 0.63
Female: 0.72
Male: 0.45
Female
Male